

BITS & PIECES

* I'm slowly loosing ground trying to keep up with information sent in to be printed and with material I have to print. I've used proportional spacing, 7 lines per inch print with a smaller type wheel, and reduction of the master to cram as much print into one issue as possible. Since I am still falling behind, I will attempt to print a 10 page issue every other month and see if that helps. If not, it will be a 10 page issue every month. Although I have done so in the past, I hate to go over 10 pages since the postage goes up from 20 cents to 37 cents to mail each issue. I also do not want to publish more than once per month if it avoidable. We'll see how 10 pages every other month works out for awhile.

* Because of the market competition, my drive prices have been substantially reduced. This will probably be the last drop I will receive, and I pass these price reductions onto you.

New prices for the Tandon TM-100-4 DS DT drives are now \$365 in 1-6 quantities. The TM-100-2 sells for \$290. The Thinline 8" Double sided drive sells for \$495. SPELLBINDER has been reduced to \$285. DBASE II remains at \$495 as does the Magnolia DD H89 setup.

* I'm presently working on plans for a 10 meg hard disk system for the H8 or H89 for under \$2000. Of course, the Tandon 10 meg hard disk will be used. If there is much interest, I may go ahead with the project. Send me your feedback. I'll keep you posted.

* Rumor has it that there is a good chance of another (beside Sextant) magazine to appear on the market for the Heath/ Zenith users. This would be a full magazine put out by one of the large publishers in the computer world. We'll wait and see what happens.

* Inflation? Not in the computer field. I have so much work I can't keep up. I've hired help, and still can't keep up. I've been looking for a larger place for some time now. Bragging? No! I'm telling you this because if you have a computer, some technical expertise or programming experience, and motivation, you can't go wrong in this advanced age. If you don't have DRIVE and motivation, forget it. If you want something to happen and want to make some extra money off your machine, or perhaps make a living, the market is there. It's not going to happen with you just sitting around wishing, but by getting involved, pushing, sweating, investing, and working long hard hours. The market is there. Take it from one who's been there, and is going strong!

Right now there is a good market for a quality 5" drive cabinet, single and double drive capacity, and horizontal mount. I have discontinued handling drive cabinets for the present time because there is a lot of junk out there, and I refuse to handle junk. Anyone who has the metal working knowledge and ideas for making a good horizontal drive design, the market is there.

Another good area is in Computer Aided Instruction (CAI). There has been very little good software out there, and one main reason is Piracy. A school will buy one copy of a good CAI program, then make 20 copies for the rest of the school. This fact was pointed out in a recent edition of one of the major computer magazines. With that kind of sale volume, what motivation does a software designer have?

Since this is along the same lines, I'll throw it in here. Just before printing time, I received a sample 2 volume set of a book by Victor Wild, "Your Fortune in the Microcomputer Business". It's published by Wildfire Publishing Company/ POB 420/ Carpinteria, CA 93013. I have no idea what the price will be. It gave me a lift to see that Victor Wild choosed to use my newsletter for a sample in his book. Starting on page 186 is the entire October 1981 issue reproduced, with some comments by yours truly. Of course, my ego trip is not why I am mentioning this book. There are many fine chapters in this two volume set with many ideas, hints, and facts about making money in this business. I highly recommend it to ALL for good reading, especially if you want to (or are) making money with your system.

* The correct address for subscribing to PASCAL NEWS is the Allentown, PA address as listed on page 2 of issue #24.

* Answer to WHO'S WHO by Richard Melvin (page 7, issue #25) on the motor spin time for UCSD PASCAL was sent in my Christopher Simmons. This was based on the HDOS article written in H8SCOOP September 1980 and modified for the Pascal by Chris. Place on BOOT disk and name 'SYSTEM.STARTUP'

```
.PROC SPIN
MVI    A,10H    ;disk spin time constant
STA    2048H    ;write to memory
XRA    A
.END
```

This routine needs to be A(ssembled) and then L(inked). Name the code file 'SYSTEM.STARTUP' and place on the BOOT disk. When Pascal is BOOTed the file SYSTEM.STARTUP will automatically be run.

* James Simpson sends this bit in. For those using a single TAPE drive with the H8-5 card there is a very simple way to enable the recorder drive motor for rewinding, etc. All you have to do is put a SPST switch across the second drive motor control lead. When you assemble the H8-5 for a single tape drive, the record and playback relays are connected in parallel. Thus if you bring out the second tape motor control lead, you can have direct control of the tape drive motor.

* More good news about the HS-013 MULTI-PURPOSE DUP/ DUMP package for \$10. The DUMP always supported a feature even I did not know about. You can DUP a 40 track disk placed in an 80 track drive TO a 40 track disk placed in a 40 TRACK drive. Thus 40 to 40 dup is possible with just one 40 Track Siemens type and an 80 track Tandon drive. Because of demand, the DUP has been modified to access any drive, up to the maximum number specified in DVDDKGEN (currently up to 6 drives with the Greg Marsh mod). It will give a 4MHz message with a 4Mhz system which it did not do before, although it did work with a 4MHz system. There is an improved disk INIT unit code to eliminate soft errors which occurred in the previous version. Another little known feature of DKDUP is its ability to install the 3 sector skewing which the UltiMeth driver uses. This skewing is installed regardless of the driver used or type of media (HDOS, CP/M, UCSD) copied. This means that users of DKDUP can realize the benefits of sector skewing on their CP/M or UCSD disks as well as their HDOS disks. The DKDUMP program will correct for tabs inserted by PIE which skewed outputs labels in earlier versions, and also can access any number of SY drives.

All versions shipped after 4/3/82 have these new programs installed. By the way, all versions of this DKDUP program need not have SELECT set to 0 as other DUP'S do. If you want the current update, send the old version disk plus \$2, or \$4 without to QUIKDATA, and you will get the current version. This has to be by far the best software package available for the price.

For those with the old release operating at 4 Mhz, simply patch 060024 from 004 to 064. This patch will allow the 4Mhz message to be printed for 4Mhz systems, as 64 is the printable value for 4, 004 is not printable.

* A note came in asking what offers will work with the version 2 HDOS 400K driver. There is nothing weird about this driver. Any program that will work with HDOS 2.0 will work with this driver, with the exception of some patches which depend on the location of certain files, which may be moved around since the cluster size of any high density drive is different than that for the standard driver. As for patches, you'll have to experiment (after making a backup of your working disk, of course) and see what works. And yes, things are becoming very confusing with all the drivers for HDOS and CP/M out. I will continue to support the DKH17V2.DVD driver I sell (from UltiMeth) and the CP/M drivers from Livingston Logic Labs. All other drivers, although they may work, will be in no way supported by myself, Dean Gibson of Ultimeth, or Ray Livingston. This may be a good time to throw in this point which many are bringing up. The HUG SY.DVD is NOT a HUG originated product, but the original SY.DVD (quite outdated at this time) which was written by Dean Gibson of Ultimeth, and originally distributed by UltiMeth and Myself.

* This HDOS CAT hint was presented to me by several individuals. If you want to see what is on the disk with all files starting with an M, for instance, you can do a CAT M*.*. This will show you all files that start with M. Roger Fennema points out that the wildcard asterisk can be used not only be itself, but with any combination of letters and numbers. Example- you may want to copy the following files on a disk; MATH1.BAS, MATH2.BAS, MATH.BAS, MATHVB.DOC and MATH6VB.TXT. The easiest way to copy these with one command is using PIP or whatever, and using MATH*.* for your command-- they will copy over. Likewise, if you were not sure what MATH files you had on the disk, you could do a CAT MATH*.*, a CAT M*.*, or a CAT MA*.* etc, to get the listing!

I also discovered that instead of COPY TT:=SY0:name.ext to get a listing on the console, you can use the same command as CP/M uses, namely TYPE name.ext (or TYPE SY1:name.ext for other than SY0:). A real time saver, and like the DIR command instead of CAT, it makes the transition between HDOS and CP/M smoother.

* Last month I listed some incorrect prices for **Hoyle & Hoyle Software** (Page 4). **Galactic Experience** is not \$18, but \$25. **Steer Kleer** is \$14. **A Remarkable Experience** is \$16. What is \$18 is the Map, Hints and Scoring guides for A Galactic Experience. Also for \$18 is the Map; Hints and Scoring Guide for A Remarkable Experience.

* A problem has come up using the STATDEF from Ultimeth, of improperly displaying drive specs. This program is set up for having SET TT: TAB. It assumes TAB is set on the Console. If it is not, set it; it is the proper way to operate it anyway, since it speeds up the console display.

* Feedback on **TECHNICAL FORUM** last month. Pat McNamara points out that AC ripple may also be a problem, especially if you built any of your own power supplies. Excessive ripple can cause some weird problems. A Scope is the best way to check this, set on

'AC'. You should observe nearly a straight line for DC power supplies, without much AC sine wave action on the line.

* During the month of MAY 1982 only, I am making available an article sent in by Fred Pospeschil on "Interfacing the Heath H-14 Printer with the H8-5 Serial I/O Cassette Board Using a RS-232 to TTL Level Converter". It was too long to print here, and I don't think the interest was high enough to warrant taking that much space, so I felt this was the best way to handle it. If anyone has the H14 and wants to run it off the H8-5, this article will tell you how to do it. Or if you just want another RS-232 port and have the H8-5 card sitting on the shelf, you may want to consider this. The article is 5 pages long. Send \$1 to cover copy costs, postage and handling. I will not honor this after the month of May.

* Fred Freeland, in response to last month's REQUEST sent me the info for operating the selectric I/O typewriter as a printer under CP/M 2.2.03 using my Selectric article project. Anybody wishing this, send me 2 disks plus \$1 shipping, copying, and handling for the .doc files and CP/M files on disk, or send \$6 if you want me to supply the disks. The BIOS has all 8" code removed to make it smaller, and the selectric routines fit where the TTY used to be.

Fred also indicated he has a modified BIOS for CP/M 2.2.02 to give it a type ahead buffer. If you are interested in getting ideas from him, write Fred Freeland/ Qtrs 2568-A Stryker/ Ft Lewis, WA 98433.

* James Kandul of **KANDUEAZY/ 6218 Blossom Ln/ Alexandria, VA 22310/ (703) 922-9450** is offering the complete line of ZDS products at a reduced price. He can supply most software & hardware (non-kit). Call or write for a complete catalog. Some items are Z25 for \$1440, MX80 for \$490, WH88-16 for \$135, Novation Modem for \$170, Hayes Smart Modem for \$250, HDOS or CP/M for \$135, MBASIC CP/M for \$155 and for HDOS \$135, and CBASIC for \$90.

* It appears that the **TRIONYX H8 Z80 CPU** is being well accepted. Joe McGlone says "Personally I think their Z80 CPU was worth waiting for. It has all the features and options one could imagine already built in. The front panel is fully supported and the 2Mhz/4Mhz switching 'on the fly' is too." I have received similar comments from purchasers of the new card, all so far, all is favorable. (See Reports)

* I have decided to take on software from reputable vendors. If you are a vendor with some good software and would like me to handle distribution, contact me. I am presently handling the software from **SOFTWARE WIZARDRY INC.**, (Tom Jorgenson). I presently have stocked **ZLYNK** and **MULTI DEVICE SPOOLER** which are \$24.95 each (includes shipping). These are both excellent programs, and were written up in NEW STUFF last month.

* Update on **4 Mhz H89 mod** as written up in December 1981 issue of H-SCOOP. If you remember, for about 10 minutes time your H89 can run at 4 Mhz. I got CP/M up OK, but was having trouble with HDOS. Robert Todd suggested I try the 2114 memory chips (located on top left of H89 CPU). This made sense, since this is the main hardware difference between HDOS and CP/M. I couldn't figure out why CP/M would run and HDOS would not. HDOS uses these 4K memory chips for a memory scratchpad, whereas CP/M does not. My 4116 memory chips in my H89 were fast enough (although I swapped them with 150ns chips to be sure), but the 2114's were not. I changed these to 150 ns chips, and all has been well ever since.

* Update on HDOS MBASIC problem (March issue) with using the underline. William Velten writes "the underline is a reserved character in HDOS (part of the editing function), and will indeed cause a backspace and rubout when used in a print statement. Thus 10 PRINT "HELLO" will cause line 10 to be blank. Since our H19/H89 uses the underline as a graphics character, we must use CHR\$(95) when it is needed".

Larry Wier wrote in some info from a Mr. Aagenas from Heath/Zenith to solve that problem. Patch MBASIC as follows, and you can then use the underline as a graphic character or whatever. Addresses are in split octal.

CHANGE locations 144254 through 144260 (old data 376, 137, 312, 124, 144) to 000. This will solve that problem.

* Peter Frank tells me that Diablo Systems, Inc has a retail store named **Diablo Systems Inc.**,/ 1510 Trimble Road/ San Jose, CA 95131/ (408) 263-7704, for those interested.

* Those interested in my **direct mail advertisement** service, I'll be getting in touch shortly. I haven't had time to work on it much yet, but your names are down.

* Bob Flocchini reports that the math chip (AMD 9511) on the FBE board and the Microtran Library make a nice combo. The port assignments have to be changed and the library has to be reassembled using the Macro-80 assembler. The library is then created using Tidewaters Library manager for HDOS. Execution time on a FORTRAN program was reduced from 68 minutes to 9 minutes.

* Chuck Lewis reports that he tried the drive support in H89 mod as described last month for placing higher density drives in the H89 and says it's worth its weight in Gold.

* Had a bad H89 CPU board from a customer. I called Milwaukee Heath and they told me replacement for that board was over \$700 and swap was over \$500. Something is wrong here. I got a brand new board, fan, power supply, etc., in my H19 to H88 upgrade kit. The entire kit, board and all was under \$400! I ended up sending it to Heath in Benton Harbor and received it back in just over a weeks time-- excellent service.

* Problems have been reported in some Magnolia DD boards for the H89. It seems several were shipped with a 10K resistor installed for R9. The value should have been 33 ohms! This was reported to have caused problems as disk errors in the middle track of the disk and upward. Changing this resistor solves the problem. Dave Bibby is the one who dug out the solution to this problem after talking to Magnolia several times. If you have this board and are having problems, check this resistor.

* Much more to print here than I had space for. I'll save it till next month.

NEW STUFF

SOFTWARE RELATED

From **Gregory Marsh**/ 5279 Miles Ct/ Woodbridge, VA 22193/ (703) 590-3360 comes **SYSTEM TESTS** for the H/Z89 and H8. It consists of three programs with source code included, to test the computer. Tests are included for the CPU chip (including seeing if it will run at higher than normal clock speeds), a FLOPYRAM test to test the 2114's that the floppy system uses, and LOWRAM to check the memory in the address range of 0 to 037.377A. The cost on this disk is \$15.

* **Christopher Simmons of Computer Systems Software**/ 2802 N Greenwood St/ La Grande, OR 97850 has a modification available that allows output for unit #6 (printer) to be sent to the H8-2 parallel board or H8-5 serial board. This mod allows the user to send output via the printer for these cards which were not used before because they were not properly initialized. To obtain this patch, send a FORMATED Pascal disk with the two files SYSTEM.HEATH and SYSTEM.FP.HEATH with \$15. If using other than port 340, specify the port you want to use so this can be configured at the same time.

* **WILLIAM DUBAY, of QUICK AND DIRTY SOFTWARE**/ 4221 Warwick Dr/ Anchorage, AK 99504/ (907) 279-3349 has available **DMS**, an HDOS data base management system which runs under 48K MBASIC. It generates sorted files of records up to 256 characters each, with utilities for editing, sorts, printouts, reports, and mail labels. Many features.

DOUBLE-ENTRY is an HDOS 48K MBASIC personal bookkeeping system for all your accounts. Designed to produce a monthly Trial Balance Statement, it allows the user to set up any number of accounts. Each transaction is entered once, debiting one account and crediting another. Utility programs include viewing and printouts of the month's ledger, a profit and loss statement, listing of all accounts and balances, and the activity of any one account for one or more months with running balances. Each of these programs is available for \$10.

* **Jim Teixeira of SOFTSHOP**/ 35 Shadow Oak Dr/ Sudbury MA 01776/ (617) 443-9693 has some new programs available under HDOS. **HFORNTH-79** is based on fig-FORTH modified to include the full set of features defined in the FORTH-79 standard. Runs under HDOS and makes use of the HDOS file structure, including named files for both program and data disk I/O. Other features include full screen editor with over 20 commands, CASE, GODO, and ARRAYS, TABLES and much more. **HFORNTH-79** may be saved on a TURNKEY basis in which it will directly enter your user program, good for commercial use. Requires H/Z89 or H8 with H19, and minimum 32K RAM. Price is \$34.95.

He also has **H25.DVD** which is a device driver for use with the H/Z25 printer. It makes extensive use of its features and allows you to configure it to look like up to 8 printers with just one device driver in memory and on disk. You can set each one for different margins, character pitch, various paper sizes, single or double spacing, overprinting for higher quality print, etc. Will automatically configure itself and send the proper escape sequence to the printer when you select the drive number (LP0: to LP7:). 15 SET options available for each of the 8 units and the HELP command shows you ALL the settings of ALL units! Similar to his popular selling **UD.DVD**. Full source code provided, and works with his **SYSMOD2** with an included patch for PR0 to PR7 commands. Requires H/Z-89 with 8250 UART (H8-4 for H8). Sells for \$29.95.

* From **Brian Lockery of INTERACTIVE MICRO SYSTEMS**/ POB 21007/ Columbus, OH 43221/ (614) 363-1534 is **RECIPE-MASTER** for \$19.95. Recipe-Master will index and select recipes from a file of recipes. The user is provided with several options which from the main menu may display one or more recipes, print them to the LP, create a sorted index, search file for specific recipes, scan recipe titles for keywords and more. Several recipe files may be present, each with up to 100 recipes. Recipes entered with any standard text editor. Supplied in .ABS format on 5" hard sector disk for HDOS 1.6 or higher.

* From **Terje Bolstad of ELEKTROKONSULT**/

Konnerudgaten 3/ N-3000 Drammen/ NORWAY comes several CP/M utility type programs. The **DISK UTILITY PACKAGE** for CP/M 2.x includes **DDUMP**, **DTEST**, **DDUP**, **DUSER**, and **UNERA**. **DDUMP** and **DTEST** were written up in **NEW STUFF** in H-SCOOP issue #22. These utilities are quite unique in that they automatically adapt to ANY diskette format. They are also the most user friendly utilities of its kind on the market, and still among the lowest in price. **DDUMP** and **DTEST** were also reviewed in Feb '82 **LIFELINES**, who missed the entire point of them being for ANY disk format. Rather than giving software such as this negative reviews, **Lifelines** should be taking time to clean up their own act first--ed.

I want to say I recently lost an entire 8" disk because the directory was unreadable (curses--more trouble with CP/M!). Naturally, I did not yet back this disk up because I was just working on it. Using **DDUP**, I was able to make a copy and recover most of the data on it. A real lifesaver!

DUSER enables you to have access from one user area to programs and/or files stored on other user areas, without having to keep duplicate copies of files on the disk. **DDUP** is a disk duplicator which is independent of disk controller, drive, disk size and format. It only requires that the source and destination disks are of the same format and density. **DTEST** tests a disk for bad spots and locks them out from later use by CP/M. **DDUMP** allows you to examine and patch any byte on any sector. **UNERA** recovers accidentally ERased files from a CP/M disk. The price for the entire set of these 5 utility programs is \$125, or \$29.95 each. Complete set of manuals \$20. Specify format 8" SS/SD or 5" H/Z disk. Add \$8 per order handling & shipping. VISA accepted.

* From Tom Jorgenson of **SOFTWARE WIZARDRY INC/** 122 Yankee Dr/ St. Charles, MO 63301/ (314) 946-1968 comes **The Multiple-Device Spooler**, based on the EasySoft Despooler. Those who purchased the original spooler package can get an update by returning your disks FREE, ONLY IF you returned the original disk with SASE suitable for the return of the disk and doc to you. You must also have a valid registration card on file with them. You will also be entitled to an additional update under the same terms in the future when such an update may become available. New purchasers of the MDS version will be charged \$10/ update.

There are many improvements in the new version. It supports ALL valid HDOS disk formats from Siemens 100K to Winchester. There are two NEW device drivers, one for the MX printers and one which supports everything from bit graphics to the H8-2 parallel card (and the H8-5 serial card). The new MDS spooler has a total of 6 command line functions. **SPOOL** (for up to 128 copies of one file), **READ** (to display file queue), **NEXT** (to abort a single queue entry), **ABORT** (to reset spooler), and **RESUME** (to restore suspended spooler operation). There are now a total of 6 device drivers supplied (for LP14, LP44, AT84, AT85, MX, plus UNIVERSAL H8-5 type driver). Includes over 20 pages doc. Price is \$24.95 plus 5% tax in Missouri. Available from **SOFTWARE WIZARDRY** and many Heath/ Zenith dealers.

* Update from **Pat McNally** as mentioned on **WORDSTAR ENHANCEMENTS** in last months issue. Pat tells me that since the majority of his requests have been for the self installing version, he is distributing that, rather than source code, for the same price. He wants to add that the package provides labels on the 25th line and enables ALL special function and keypad keys. Control keys will still be needed, however. He also has a **WORDSTAR** mod package making use of the special functions of the MX-80, which is also self installing and sells for \$19.

Also available from Pat, is **GRAPH-IT** for \$29 which is a versatile graphing system to plot data from SuperCalc, Condor, dBASE II, BASIC or almost any program. This is for CP/M. Use ANY printer to plot up to 50 items or functions simultaneously, for any number of periods. Graphics can be created or edited easily. Many features.

FLIGHT SIMULATOR for \$19 is a simple simulator, and not a game. It's tough and takes some time to learn and there's no 'canned flight plan' to follow. Using your full screen instrument panel mock-up, fly where you want and when you want, in a glider, trainer, or 747. You define runway and wind conditions, so when you get better the going gets rougher. Fast, accurate and challenging. Requires CP/M 2.2. All programs distributed on 5" hard sector SD disk. Add \$2 per order for postage and handling. For those not here last month, address is POB 578/ Haleiwa, HI 96712.

* Update from Ron Rocheleau of **NEWLINE SOFTWARE/** POB 402/ Littleton, MA 01460/ (617) 486-8535. Quite some time back I printed some info and feedback on **VIDEO SCRIBE**, an HDOS word processing package (yes, it formats too). I have received much positive feedback on it. At that time the major item I wrote against was the bad spacing produced from right justification. That's now fixed, and the new version with extra features (wrap around, built in HELP command, centering of text, OOPS undo mistake key, printer control escape functions can be sent, and more) is available for \$39.95. One of the better word processors under HDOS around, a good buy for the money.

They also have for \$39.95 **VIDEO ARTIST PLUS**, which is a graphics editor, and **TOUCH TYPIST**, for \$29.95 (HDOS and CP/M version available) to teach you touch typing (or help you improve). New products released by them are **H25/Z25 GRAPHICS GENERATOR** for \$44.95. This program contains all the features of **VIDEO ARTIST PLUS**, for the creation of pictures on the H19 terminal to be saved to disk or printed to H25 printer. Screens can be combined to print 160 column wide by any length pictures! Include other printer commands. Requires HDOS and 32K memory.

TOUR.700 is another new program for \$19.95 and available in HDOS or CP/M. This is a 'different' kind of card game to amuse you without tiring you. A good mix of skill and luck of the cards. The H19 graphics present a board display. The object is to win a cross country race against the computer. You must overcome flat tires, accidents, speed limits, and other hazards as you make your way through each 700 mile leg of the track (sounds like a familiar card game, forget the name-ed). HDOS version includes both **BHBASIC** and **MBASIC** code, the CP/M version with **MBASIC** only (CP/M 2.x or higher).

* Something unique from **Ralph Avery** of **STAN AVERY & ASSOCIATES INC/** POB J/ Boca Raton, FL 33432/ (305) 368-0170. **INSURANCE AGENCY ACCOUNTING PACKAGE** is a CP/M software package under Microsoft COBOL. It is an interactive sophisticated data entry system developed to make operation easy. Auto invoicing is the heart of the system, which is linked together with A/P, A/R, Cash Receipts Journal, Cash Disbursed Journal, G/L, Profit & Loss, Customer/Label, and Mailing List. This complete package starts at \$2500 and can be tailored to your additional requirements, and can grow with you. The price includes 2 full 8 hour days of on site training at your agency. Problems are handled on site or by phone.

* From **Bruce Bennett** of **MOUNTAIN SOFTWARE CO/** 299 Smith Hill Rd/ Utica NY 13502/ (no phone) is **STARCAT.COM** which operates under CP/M. It sells for \$29.95 and is supplied on 5" hard sector floppy, or for

\$36.95 on 8" CP/M standard disk. This is a menu driven, but complex program for amateur astronomers. It calculates the sidereal time and displays data for celestial objects which are in the user's area sky (which is defined by you). Files may be retrieved from any or all of the following: Naked eye stars, Binary stars, Variable stars, Messier objects, Galactic nebula, Globular clusters, Open clusters, and Galaxies. The included data base contains more celestial objects than the average amateur will take the time to observe. The user may expand and/or modify any of the data files to accommodate his own preference. Many, many extra features. Requires 48K and CP/M, dual disks, 80 character video terminal, and optionally 80 character printer. Write Mountain Software for full info.

* Software from **Livingston Logic Labs/** 2386-323 E Del Mar Blvd/ Pasadena, CA 91107/ (213) 792-4763 will be available to allow the Heath/Zenith CP/M to support the Magnolia 128K RAM board. These are similar to the software products available from UltiMeth for HDOS as written up last month. **INVDISK.COM** is a program which patches itself into memory above the BIOS and links itself into the BIOS to provide a 112K logical disk. This allows the memory to be read from and written to in exactly the same manner as a disk drive, except with MUCH faster access. **CACHE22.COM** also patches itself into memory above the BIOS to provide a 'cache' memory. This is used to reduce the number of physical disk accesses by storing the most frequently read sectors in the cache memory and retrieving them from memory rather than disk whenever possible. In many cases disk access can be almost eliminated, with an enormous increase in execution speed. The price of this package is \$40.

Also from Ray Livingston is the HDOS disk driver for the CDR systems FDC-880H DD 8"/5" soft sector disk controller for the H89. This driver is fully compatible with the H37 and H47, and the UltiMeth driver for the Magnolia DD controller. It will support up to 4 drives in any combination. It will also handle both 40 TK and 80 TK single and double sided 5" drives, as well as the new (not yet released) 154 Track 8" drives. The price on this is \$40.

* Several products are available from **Extended Technology Systems/** 1121 Briarwood/ Bensalem, PA 19020/ (215) 754-4604. **ETS-C2.2 ZCPR** is a Z80 replacement for the CCP in your CP/M operating system. It includes all normal commands plus several HDOS oriented commands such as CAT and DEL. There are many other features, write or call for complete info. This is an adaptation of SIG/M Volume 54 to the Heath environment. Price for ETS-C2.2 ZCPR is \$20 which includes 40 TK disk with source and several .HEX files. Also included are several SUBMIT files which will build and patch this into your system on both 40 and 80 track disks, for Heath 2.2.02, 2.2.03, or Magnolia 2.2 CP/M. When ordering specifying which of these CP/M's you have.

ETS-H2.1 is a new SYSCMD.SYS for HDOS 2.0. This HDOS command processor represents as much an enhancement beyond the current enhancements of HDOS such as the Teixeira SYSCMD, as Teixeira's was beyond the standard HDOS. Final product to sell for about \$50, preliminary version available now, for \$40. Owners of preliminary will be allowed a free update when final version is ready. See Hardware Related items for more from ETS.

HARDWARE RELATED

* From Don Long of **ENVIRONMENTAL CONTROL SYSTEMS/** 9319 Willowview Ln/ Houston, TX 77080/

(713) 464-1717 is their latest **PC-89**, a 9 port Parallel I/O interface card with perpetual Clock/Calendar for the H/Z89. Includes extensive documentation and software \$224.95. This was born from the PC-12 for \$249 which was the 12 port version for the H8. If you want to inspect documentation before you buy, PC-DOC for \$15 is the manual set for the PC-12A and PC-89. The price is applicable toward future purchase. **RX-10** is a Universal BSR X-10 Remote Control System Computer Interface for the introductory price of \$124.95. **RL-10** is a two channel (Stereo) Remote Audio Level Controller for the introductory price of \$49.95.

They also sell support cables, using low profile 14 pin DIP plugs with heavy duty gold pins. A one time \$6.50 discount applies with the purchase of a PC-12A or PC-89 or RX-10. A CB2 cable will directly connect an RX-10 to a PC-12A or PC-89. For custom cable lengths, contact them. **CB1-5** Single end 5' cable \$6.50. **CB2-5** is a double end 5' cable for \$8.50.

* From Ted Benglen of **MI-8/** 822 E Country Rd 30/ Ft Collins, CO/ (303) 669-4116 comes a variety of new products. They are also handling the assembled and tested PC-89 with real time clock and 9 parallel ports for \$224.95.

COMMAND-8 is the **VOICE RECOGNITION** and **CONTROL SYSTEM** from MI-8. Has the accuracy and versatility of voice recognitions costing far more. Allows you to add voice control to games, command your computer to turn off and on lights and appliances, or load and run a program, and more. Comes assembled and tested with doc and support software on disk. Software includes several demo programs using the BSR home controller and V-8 Voice Synthesizer for voice response, a TEST program, and a practice recognition program. With **COMMAND-8** you will be talking to your computer in no time, and it will be listening (with V-8 it will also be talking back!--ed). System requirements are Z80 processor, parallel input port and 48K RAM. Intro price is \$99.95 through June '82, when the price will then be \$124.95. Don't let this low price fool you, **COMMAND-8** really works, and they guarantee it!

DIRECT CONNECT MODEM for \$99.50 is a package of a direct connect modem hardware pack and software. Like any standard Modem, it operates at 300 baud in full duplex, and has auto detect. A LED indicates when the MODEM has locked on to its host computer and is ready for communication. Merely log on and you're ready to access the BB's. The communication software includes many user functions such as auto log on and file transfer.

They also handle the **MX-80 printer** for \$474.95, shipped from stock within 48 hours. It can be parallel interfaced with the PC-89 for an additional \$65 serial board savings!. For more info on any of these items, or to order, contact MI-8. Include 5% shipping and insurance, VISA, MC, Check accepted.

* Dave Brockman of **FBE RESEARCH CO, INC/** Box 68234/ Seattle, WA 98168 (206) 655-3711 (days) wishes to announce that the **H89RAM** 16K add on memory board as mentioned last month will not be sold without memory chips. He has to place the chips in to test it, so feels it is in everybody's best interest to leave em in. The new price with chips, assembled and tested is \$64.95. This includes a spare memory chip, and a spare U561 (74L132).

As long as we're on H89 memory, **Extended Technology Systems** (see Software) has available the **ETS-64** which is a 16K memory expansion for the H89 to bring it up to 64K. The unique thing about this is it uses no memory expansion slots. The kit uses a new double decker 16 pin

socket. Introductory price for H-SCOOP subscribers is \$40 for kit price without the memory chips. This offer good till July 1, '82. Normal price will be \$49 without 4116 chips and \$65 with 8 4116 chips.

And yet another 16K memory expansion board available from Joe Williams/ Rt 2 Box 207E/ Buchanan, VA 24066/ (703) 254-1686 evenings, or 563-0371 X 426 days. This is electrically similar to the one Heath/ Zenith sells, but available for \$21 for bare board and detailed documentation, with pictures and diagrams.

* Update from RAY ALBREKTSON/ 19 W Dover Dr/ San Bernardino, CA 92407/ (714) 882-8072. He wants to make note that his H89 Clock/Sound boards are now available assembled and tested for \$90 which includes much software on 5" HDOS hard sector disk. He offers a NO STRINGS guarantee and IMMEDIATE DELIVERY-off the shelf stock!

* A Preliminary Announcement of the H-89-3 COLOR GRAPHICS BOARD from Dave Troendle of New Orleans General Data Services, Inc./ 7230 Chadbourne Dr/ New Orleans, LA 70126/ (504) 241-9495. They are starting production on their color board for the H89/Z90 computers. It will display 16 colors with a resolution of 192 X 256 dots using the TMS-9918A Video Display Processor. Other board features will include: AY-3-8910, 3 channel Programmable Sound Generator with two 8 bit parallel ports; ADC-0808 8 channel 8 bit A/D Converter; 8253 triple 16 bit Counter- Timer; and 8259 Priority Interrupt Controller. In addition, the following options will be available: 9511A Floating-Point Math Chip; Two AD7542 12 Bit D/A Converters with a precision reference; and SC-01A Vortax Speech Synthesizer. The board must plug into the right slots of the CPU and has fully buffered address, data and control lines. Completely compatible with Digital Research's PASCAL MT/+, which is soon to be released by New Orleans General Data Services (NOGDS). Board to be available for \$399 assembled & tested in June, with doc and software (CP/M, HDOS, all H-17/37/47 formats). Write for more info and quantity discounts.

* Ray Livingston also announces that he will soon have available several multi-purpose I/O boards for the H89. One to replace the Heath 3 port serial I/O board will contain 3 serial ports, 3 parallel ports, and either an H67 interface or a real time clock. There's a possibility of a board for the Left side of the CPU to support a 16K RAM expansion, 3 parallel ports, and a real time clock. There's more in the works, more info will be printed as these become available.

* One item we saw while at Heath was a self contained information terminal capable of communicating with BB's and other modem units operating at 300 baud. This terminal, the ZT-1, connects to a phone line with a modular jack and is capable of auto dial to gain access to the computer and request info. The ZT-1 displays choices available from power up, which leads to other menus. Telephone numbers and passwords (up to 26-16 digit numbers) need only be entered once, and can then be used by selecting the letter of the alphabet from the menu that matches that number. The electronics are contained in a 63 key data entry keyboard. For a display it uses the Zenith ZVM-121 video monitor (included). Baud rates are selectable from 110 to 2400 for the serial I/O printer port. Parallel centronics type printer interface is also standard. Suggested price is \$695

MISCELLANEOUS ITEMS OF INTEREST

* From one of the GOOD GUYS, Ray Massa of THE KEYBOARD STUDIO/ 125 Aspen/ Birmingham, MI 48009/ (313) 645-5356 is their newest catalog #282, free on request. This full 12 page catalog features all their

present software (over 19 different products!), diskettes, fans, the H89 buss extender board, many different types of disk drive enclosures with power supplies (including for the 8" thinline Tandon), the LLL 8" D/S SD controller for the H8 and H89, Modems, Tandon disk drives of all vintages, Type 'N' Talk voice synthesizer for RS 2323 serial port, HOUSEMASTER the H89 multi function controller set, professional software, and much more.

* Ralph Bradburn of SPEED O PRINTS/ 300 Route 59/ Tallman, NY 10982/ (914) 357-2223 has various sundry items available. He can standard 3.5" X 15/16" mailing labels in continuous tractor stock of 2 and 3 across. 500 to 5000 are .0045 each, and .0035 for greater quantities. Box of 10,000 2 across is \$29.50 (less than 1/2 of local store prices), while box of 15,000 3 across is \$44. 2 across on 8" carrier, while 3 across is supplied on 11.5" carrier. Add Shipping to all these prices. They also have available computer forms, including invoices, payroll checks, A/P checks, and statements. Contact them for more info.

Ralph also points out that the HUGNJ newsletter is available for those interested in exchanging newsletters or subscribing. It is a New Jersey local HUG newsletter.

* A local salesman stopped by my office recently with all kinds of good info on forms (and samples also). It turns out this guy lives a few houses down from me, and he represents DELTA FORMS. If anyone wants custom continuous forms, contact J. Simenz/ 716 Wilson Ave/ Sheboygan, WI 53081/ (414) 452-9104. I have very impressive samples here of Checks, Invoice statements, A/R and A/P forms, True BOND letter quality continuous form paper (with so many tear perforations when you tear the tractor feed off you can't hardly see it was there), green bar continuous form 15" paper (much superior to quality you find in most stores), one piece envelope message notice continuous forms and more. You name it, if it's custom and continuous form, he can have it made up at a fair cost. Call or write with your needs, and tell them QUIKDATA sent you.

REPORT

TRIONYX Z80 CPU

This was sent in by Jim Benesch who recently built the Trionyx Z80 CPU board.

"After months and months of waiting and gentle probing, my Trionyx Z80 board kit finally arrived. An examination of the documentation and the board explained the wait. Trionyx had not only designed an innovative Z80 CPU board, they had empirically lab and field tested their design until they had a product as reliable as their memory and mother board.

"I assembled the Trionyx Z80 CPU kit in approximately 12 hours. By following the construction directions exactly, it worked the first time. Documentation is excellent. To assemble the kit, a wire wrapping tool is required since there are 30 jumpers for the standard configuration. I found two minor problems. The resistor leads were dirty and the pads for the wirewrap pins were small, but the kit's instructions advise the builder how to compensate.

"The best of its many innovative features were the programmable clock and the hex front panel display and keypad. Software needed to control the clock is simple to use and supplied free with the board on a dual mode (CP/M and HDOS) disk. For HDOS, there is a speed utility which changes the clock speed as a system

command and a prologue program to change the speed when booting. New timing constants are provided for a 4 mHz SY.DVD, but I found that Dean Gibson's 400K DKH17V2 SY driver automatically checks to see if the processor is running at two or four mHz and adjusts itself. For CP/M, there are similar clock utilities for 4 mHz. Since I am running a Tandon 400K drive, I will be implementing Ray Livingston's CP/M mods listed in the last issue of H-SCOOP. (NOTE: The latest releases of Livingston's BIOS-80 for .03 CP/M has both the 2 and 4 Mhz versions already installed, you simply pick the one you want--ed).

"The most exciting and "friendly" feature of the whole package is Trionyx's new FWZ80 ROM. It fully supports XCON and PAM ROM's but adds a hex front panel default mode. Binary I/O and a special display for flags are useful new features. "The new ROM handles the full Z80 prime register set. A powerful debugging function indirectly displays the contents of a register and the contents of the memory pointed to by that register. Ports can be continuously monitored.

"There are three DIP switches on the board. The first switch allows the CPU to run at 4 mHz with any H8 memory by adding wait states. Worst case is two wait states that runs bench mark programs 30% faster than 2 mHz. The second switch selects the enable address for ROM 1. The third switch provides the same features as the old ORG Zero board (no longer needed). There are 118 wirewrap pins on the board which provide interrupt configuration, processor status port, bus assignments and the three basic ROM options (FWZ80, XCON8, or PAM). In addition, they allow clock refinements, memory write timing, side select level, 2 or 4 mHz powerup, and single step at 2 or 2/4 mHz.

"The H8 keypad takes on a new look with the shiny plastic labels identifying the hex and function assignments. One button boot is now confined to the "GO" key while primary and secondary boot requires a two keystroke sequence. In all, I find the new H8 front panel implementation much easier to use and understand. The Trionyx Z80 board, mother board, and 64K memory in an H8 make it the most powerful and reliable 8 bit microcomputer on the market. It was definitely worth waiting for."

REPORT ON H89 NOGDS COLOR BD & MT+ PASCA

This info sent in by Fred Pospeschil and is printed practically verbatim. Since NOGDS has played an important role in the H8 product development with the music and graphics board, I felt they deserved this mention. Since Dave is working on the MT+ Pascal, and I get lot of questions on that, I felt the later part of this article also deserved printing--ed.

"I recently had the opportunity to spend some time with Dave and Jean Troendle, John Souvestre, and Roy Jenevein - otherwise known as New Orleans General Data Services, Inc. (NOGDS). These are the people who brought us the music and color boards for the H8. After seeing how they assemble and test their products I can understand why my graphics and music boards work so perfectly. Each and every board is visually inspected for construction errors and then burned in for 100 hours. After burn-in, the boards are tested and exercised in two and four-meg systems. As a result of this testing the board return rate is almost zero.

"If you thought the H8 graphics board looked and ran good wait till you see the H89 version. During this visit I was able to see the PC board layout which is close to being finished. The board is the same size as the three-port I/O board. Although its smaller than the H8 version, its functionality is greater. It has the TMS9918A

graphics and GI AY3-890 three channel sound chip, eight analog-to-digital input channels, two 8-bit parallel input/output ports, 16K of graphics refresh memory, and the socket for the AMD or INTEL math chip. In addition, it has a priority interrupt controller, and a three channel counter/timer as well as sockets for two 12 bit digital-to-analog output ports and another socket for the VORTAX synthesizer.

"I was particularly pleased to see that the analog-to-digital channels and parallel I/O ports were fully implemented. Without these facilities most graphics programs tend to be somewhat crude and the human interface is inefficient and, at times, awkward. These hardware facilities also make writing graphics software more straight forward (which Fred does--ed) and enjoyable. More about graphics software later. As of this time it looks like the initial boards should be available around June. The boards will probably be directly marketed by NOGDS. Drop Dave a note if you want to be one of the first to get one of these boards.

On the software side, Dave is making good progress with the HDOS implementation of Pascal/MT+. Beta test is well along and the system is looking fine. Dave has streamlined and enhanced several aspects of the system so it is even better than the CP/M version reviewed in the March issue of BYTE. Some of the delay in getting this product to market was with Digital Research knowing a good thing when they saw it. Midway through the conversion they bought MT MicroSYSTEMS and Pascal/MT+. That's why the HDOS version dropped out of the MT MicroSYSTEMS ads. In response to previous comments about their documentation, the manual is now considerably easier to use and greatly expanded. It is now well over 200 pages. It now has an extensive cross reference and a lengthy appendix which covers file formats, I/O, and handling. Beta testing MT+ has been one of the most enjoyable and rewarding projects I have been involved with. It looks like NOGDS will also be marketing this products for around \$350 (same as CP/M version). It should be out before the color graphics board. There are a number of good Pascals available, see my contribution to the PASCAL CORNER in the March REMark, but for my money MT+ lets me get maximum performance from my H8/H19. For raw speed, flexible programming support, and easy interface with MACRO80 it's great. For another viewpoint on MT+'s speed see Alfred French's "SPEED ON HEATH MICROS", also in the March REMark. The first HDOS release will be version 5.5 which is the same as the latest CP/M version.

As long as I am talking about forthcoming products I might as well mention the graphics support package I am putting together with considerable help from Dave. After using the graphics board for over a year I found that it can usually handle data faster than the CPU can generate it. This is especially true when plotting points and drawing lines in the G2 (192 X 256) mode. Since the applications I plan to write require a lot of trig and line drawing I can use as much speed as I can get. MT+, with or without the math chip, gives me the best trig speed. Although this support package will provide control and use of all the board functions, it's main strength, thanks to Dave, is a very fast line drawing routine. At present it is putting out about 4,500 pixels per second regardless of line direction. The package will also provide line erasing and toggling and the same for points. Other useful composite functions, such as rubberband lines, are being designed. Initially, these routines will be callable from MT+ and MACRO80. With any luck, they may also be packaged for use with Tiny Pascal. FORTRAN is also possible if enough people are interested. So far, all the graphics programs I have seen have been written in ASM or Pascal although I have heard of one good one written in FORTRAN. More later when the package firms up.

All in all, it was a most enjoyable and educational visit. I hope that these tidbits are of use in keeping you informed on some of the things which are going on in the community. If you want more info on this product, drop me a SASE at 3108 Jackson St/ Bellevue, NE 68005. I will return an overview when it is completed.--Fred Pospeschil (402) 291-0795 between 7 and 9PM central time.

CLASSIFIEDS

FOR SALE--TM-100-2, 40 track double sided disk drive for \$225, or will consider a TRADE for a used bare H8. With the new SY driver, this will do nicely as a 200K single density (400K DD) drive. Also 64K worth of 4116 NEC memory chips for \$1/K (set of 8 chips for \$16). Jim Benesch/ Box 1811/ Minden, NV 89423/ (702) 266-3686.

FOR SALE--16K Heath static RAM board (2) for \$150 each. Heath 8K static RAM boards (4) for \$50 each or 2 for \$90. Chris Simmons/ 2802 N Greenwood St/ La Grande, OR 97850/ (503) 963-4845.

FOR SALE--H89 with 64K RAM, ROM kit, H14, one Tandon drive with much software included. \$2,200. Contact Michael Steiner/ 3044 Sunset Ave/ Rocky Mt, NC 27801/ (919) 443-6129.

FOR SALE--H-14 printer, \$300 or best offer. Upgraded for H25. Budget Print Center/ 571 Valley Rd/ Upper Montclair, NJ 07043/ (201) 744-5520 days, 6308 night.

WANTED--EPSON 8114 RS 232 serial I/O board for the MX-80. Will pay \$50. Used Assembler Language course wanted--Cheap!. J Czebiniak/ 199 Vly Road Extension/ Schenectady, NY 12309/ (518) 869-8723.

FOR SALE--Four WH8-16 RAM boards for \$150 each. Richard Tawes/ 148 Oak St/ Bridgewater, NJ 08807/ (201) 725-9697 after 6PM, or 822-2100 X 2323 from 9-5 (work number).

FOR SALE--H8 computer with 8080 CPU for \$200. Trionyx 64K memory board for \$300, WH-8-4 4 port serial board for \$125, HA-8-8 Extended Configuration Board for \$25, H8-5 Serial and cassette controller for \$30, H-17 disk controller board (board only) for \$65. Will sell the entire above package for \$625. Also have an H88-5 Cassette interface for the H88 or H89 for \$25. Thomas Bessire/ 7600 Pacific Place Ft Worth, TX 76112/ (817) 451-3912.

FOR SALE--H-14 printer. \$275 plus shipping. Bill Bechtold/ 7429 Frederick St/ Omaha, NE 68124/ (402) 397-2461.

Two Siemens FDD-100-5B004 floppy 5" disk drives, single track, single side, in mint condition for \$250 each. Hank Guell/ 246 Skyline Lk Dr/ Ringwood, NJ 07456/ (201) 839-9071 after 6PM EST.

FOR SALE--Bare H8 without memory or I/O boards for \$200. H8-4 four port serial board for \$125. H8-5 Serial Cassette I/O board for \$60. Trionyx 64K memory (unmodified) \$275. H-17 disk controller board for \$75. Programmable Sound Generator Board for \$20. All of the listed items for \$700, and an assortment of cables and software included. Manuals included with all. John Bennin/ Rt 3, 620 Hill St/ Baraboo, WI 53913.

FOR SALE--New H89 green screen system with 64K RAM, dual H-37 drives with both disk controllers included, H-25 printer, Extra diskettes, and much extra software including a complete business package. Mark Riemer/ 293 Union Dr/ Rochester, MI 48063/ (313) 857-3138 (work hours) or 652-0381 (home).

WANTED--H-10 Paper tape reader/ punch. Also wanted H-17 box and power supply (don't need drives or interface card, but will consider them if a package deal is all I can get). **FOR SALE**--H-8 with 8080 CPU. Make offer. Louis De Michele/ Unit AA/ POB 561/ APO NY 09098.

FOR SALE--H 14 Printer. \$275 plus shipping. Contact Bill Bechtold/ 7429 Frederick St/ Omaha, NE 68124/ (402) 397-2461.

FOR SALE--H8/H9/H17 computer with Trionyx 64K memory. H9 has Northwest Mods for cursor control, 24 line, and 4800 baud. \$1000. Steve Hillman (906) 635-1052- days, or 632-3883 evenings.

FOR SALE--H8 computer complete with 2 8K memory boards, H8-5 serial/ cassette board, and disk controller board (board only no drives). All in excellent condition. \$350. Milt Krauthoff/ N89 W16800 Cleveland Ave/ Menomonee Falls, WI 53051/ (414) 251-2155.

WANTED--H17 with high quality 2 or 3 drive power supply and case, but without drives and controller card! All I am looking for is a quality cabinet with power supply for 2 drives. Parnell Lewis/ 199 Cherokee Dr/ Memphis, TN 38111/ (901) 454-0783.

FOR SALE--H14 printer for \$300 or best offer (moved up to H25). Robert Evans/ 571 Valley Rd/ Upper Montclair, NJ 07043/ (201) 744-5520 days, 6308 nights.

FOR SALE--WH8-16 16K RAM (5 of these) in mint condition, 3 brand new! \$125 each. Two 8K RAM in excellent condition for \$50 each. Z89 16K upgrade kit, brand new and in factory box, \$125 each (2 of these).

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PHONE HOURS: 9AM to 9PM as I am available, except sunset Friday to sunset Saturday (Seventh Day Adventist). NO COLLECT CALLS.